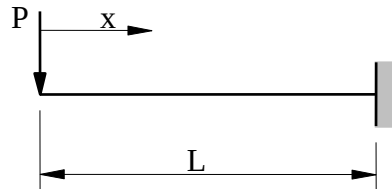


THE UNIVERSITY OF BIRMINGHAM
DEPARTMENT OF CIVIL ENGINEERING

Mechanics 1

Tutorial Sheet - 5

1a



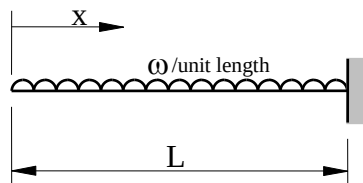
$$EI \frac{d^2 v}{dx^2} = -M = Px$$

$$EI \frac{dv}{dx} = P \frac{x^2}{2} + A \quad @ \ x=L, \ dv/dx = 0. \ \therefore A = -PL^2/2$$

$$EIv = P \frac{x^3}{6} - \frac{PL^2}{2}x + B \quad @ \ x=L, \ v = 0. \ \therefore B = PL^3/3$$

$$@ \ x = 0 \quad v = \frac{PL^3}{3EI} \quad \& \quad \frac{dv}{dx} = \frac{PL^2}{2EI}$$

1b



$$EI \frac{d^2 v}{dx^2} = -M = \frac{\omega x^2}{2}$$

$$EI \frac{dv}{dx} = \frac{\omega x^3}{6} + A \quad @ \ x=L, \ dv/dx = 0. \ \therefore A = -\omega L^3/6$$

$$EIv = \frac{\omega x^4}{24} - \frac{\omega L^3 x}{6} + B \quad @ \ x=L, \ v = 0. \ \therefore B = \omega L^4/8$$

$$@ \ x = 0 \quad v = \frac{\omega L^4}{8EI} \quad \& \quad \frac{dv}{dx} = -\frac{\omega L^3}{6EI}$$

-
- 2 Vertical reactions: LHS = $\omega L/6$; RHS = $\omega L/3$
(NB value of load at RHS = ω per unit length. Hence value of load at any distance x from the left hand support = $\omega x/L$)

$$EI \frac{d^2v}{dx^2} = -M = -\frac{\omega Lx}{6} + \left(\frac{1}{2} \frac{\omega x}{L} x \right) \frac{x}{3} = -\frac{\omega Lx}{6} + \frac{\omega x^3}{6L}$$

$$EI \frac{dv}{dx} = -\frac{\omega Lx^2}{12} + \frac{\omega x^4}{24L} + A$$

$$EIv = -\frac{\omega Lx^3}{36} + \frac{\omega x^5}{120L} + Ax + B @ x=0, v=0. \therefore B=0$$

$$@ x=L, v=0. \quad AL = \frac{\omega L^4}{36} - \frac{\omega L^4}{120} = 7 \frac{\omega L^4}{360} \quad \Rightarrow \quad A = 7 \frac{\omega L^3}{360}$$

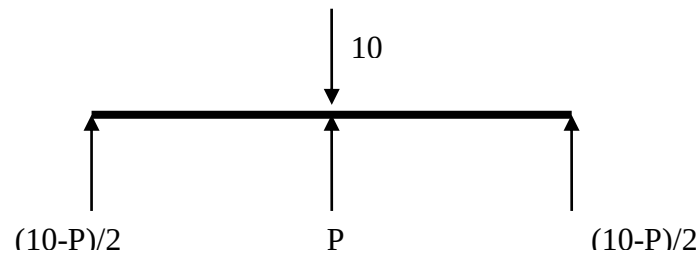
$$\text{For maximum deflection, } dv/dx = 0: \quad \frac{\omega x^4}{24L} - \frac{\omega Lx^2}{12} + \frac{\omega L^3}{360} = 0$$

$$\Rightarrow 15 \left[\frac{x}{L} \right]^4 - 30 \left[\frac{x}{L} \right]^2 + 7 = 0 \quad \therefore \left[\frac{x}{L} \right]^2 = 0.270 \quad x/L = \mathbf{0.519}$$

At which point:

$$EIv_{\max} = -\frac{\omega L^4 [0.519]^3}{36} + \frac{\omega L^4 [0.519]^5}{120} + -\frac{7\omega L^4 [0.519]}{360} = 0.00652\omega L^4$$

3 Consider upper the beam:



where P = the reaction force between the upper and lower beam.

$$EI \frac{d^2v}{dx^2} = -M = -\frac{[10 - P]}{2}x$$

$$EI \frac{dv}{dx} = -\frac{[10 - P]}{2} \frac{x^2}{2} + A = -[10 - P] \frac{x^2}{4} + A \quad \text{when } x = L/2 = 4, dv/dx = 0.$$

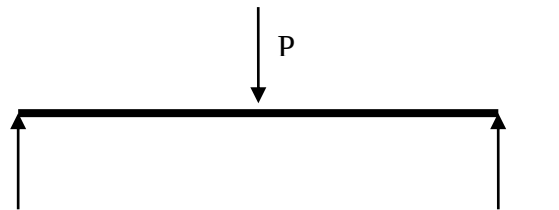
$$\therefore A = (10-P)x16/4 = 4(10 - P)$$

$$EI \frac{dv}{dx} = -[10 - P] \frac{x^2}{4} + 4[10 - P]$$

$$EIv = -[10 - P] \frac{x^3}{12} + 4x[10 - P] + B \quad @ \ x=0, v=0 \therefore B = 0$$

$$\text{at } x = L/2 \ v = v_{\max}: \quad EIv_{\max} = [10 - P] \left[16 - \frac{64}{12} \right] = \frac{32}{2} [10 - P] \quad [1]$$

Consider lower beam:



$$EI \frac{d^2v}{dx^2} = -M = -\frac{P}{2}x \quad \Rightarrow \quad EI \frac{dv}{dx} = -P \frac{x^2}{4} + A$$

When $x = 3$, $dv/dx = 0$, hence $A = 9P/4$

$$\therefore EIv = -P \frac{x^2}{4} + \frac{9P}{4}x + B \quad @ \ x=0, v = 0, \text{ hence } B = 0$$

Maximum deflection of lower beam occurs when $x = 3m$, hence:

$$EIv_{\max} = -P \frac{27}{12} + \frac{27}{4}P = \frac{27P}{6} \quad [2]$$

Since the deflections of the two beams must be identical it follows that $[1] = [2]$.

Hence,

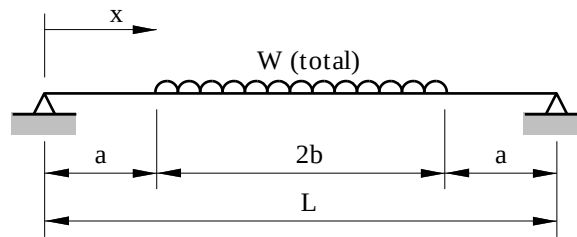
$$\frac{27}{6}P = \frac{32}{2}[10 - P] \quad \Rightarrow P = 7.03 \text{ kN}$$

Greatest bending moments:

Upper beam = $(10 - P) \times 4/2 = 5.93 \text{ kNm}$

Lower beam = $(P) \times 3/2 = 10.55 \text{ kNm}$

4



$$M = \frac{W}{2}x - \frac{W}{2b} \frac{< x - a >^2}{2} \quad (0 \leq x \leq L/2)$$

$$EI \frac{d^2v}{dx^2} = -M = -\frac{W}{2}x + \frac{W}{4b} < x - a >^2$$

$$EI \frac{dv}{dx} = -\frac{W}{4}x^2 + \frac{W}{12b} < x - a >^3 + A$$

$$@ x = L/2, dv/dx = 0 \text{ (symmetry)} \quad A = \frac{WL^2}{16} - \frac{W}{12b}b^3 = W \left[\frac{L^2}{16} - \frac{b^2}{12} \right]$$

$$EIv = -\frac{W}{12}x^3 + \frac{W}{48b} < x - a >^4 + W \left[\frac{L^2}{16} - \frac{b^2}{12} \right] x + B$$

At $x = 0, v = 0$, hence $B = 0$

$$v = \frac{W}{EI} \left[-\frac{x^3}{12} + \frac{< x - a >^4}{48b} + \left[\frac{L^2}{16} - \frac{b^2}{12} \right] x \right]$$

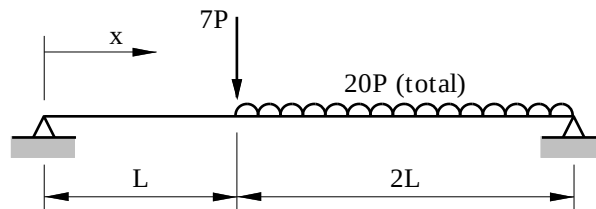
@ $x = L/2$ (NB $L/2 = a + b$)

$$v_{\max} = \frac{W}{EI} \left[-\frac{L^3}{96} + \frac{b^3}{48} + \frac{L^3}{32} - \frac{b^2 L}{24} \right] = \frac{W}{48EI} \left[L^3 - 2b^2 L + b^3 \right]$$

$$\text{When } b = L/2 \quad v_{\max} = \frac{W}{48EI} \left[L^3 - 2\frac{L^3}{4} + \frac{L^3}{8} \right] = \frac{5W}{384EI}$$

$$\text{When } b = 0 \quad v_{\max} = \frac{WL^3}{48EI}$$

5



$$V_1 + V_2 = 27W$$

$$V_1 3L - 7W \cdot 2L - 20WL = 0; \quad V_1 = 34W/3 \quad \& \quad V_2 = 47W/3$$

$$M = \frac{34P}{3}x - 7P \langle x - L \rangle - \frac{20P}{4L} \langle x - L \rangle^2$$

$$EI \frac{d^2 v}{dx^2} = -M = -\frac{34P}{3}x + 7P \langle x - L \rangle + \frac{10P}{2L} \langle x - L \rangle^2$$

Integrate twice

$$EI \frac{dv}{dx} = -\frac{34P}{6}x^2 + \frac{7P}{2} \langle x - L \rangle^2 + \frac{10P}{6L} \langle x - L \rangle^3 + A$$

$$EI v = -\frac{34P}{18}x^3 + \frac{7P}{6} \langle x - L \rangle^3 + \frac{10P}{24L} \langle x - L \rangle^4 + Ax + B$$

@ $x = 0$, $v = 0$, therefore $B = 0$

$$x = 3L, v = 0, \therefore A(3L) = PL^3 \left[\frac{34 \times 27}{18} - \frac{7 \times 8}{6} - \frac{10 \times 16}{24} \right] = \frac{105}{3} PL^3$$

$$A = \frac{35PL^2}{3}$$

$$EI v = -\frac{34P}{18}x^3 + \frac{7P}{6} \langle x - L \rangle^3 + \frac{10P}{24L} \langle x - L \rangle^4 + \frac{35PL^2}{3}x$$

$$EI \frac{dv}{dx} = -\frac{34P}{6}x^2 + \frac{7P}{2} \langle x - L \rangle^2 + \frac{10P}{6L} \langle x - L \rangle^3 + \frac{35PL^2}{3}$$

From the equation of the deflection slope

$$EI \frac{dv}{dx} = -\frac{34P}{6}x^2 + \frac{7P}{2} \langle x - L \rangle^2 + \frac{10P}{6L} \langle x - L \rangle^3 + \frac{35PL^2}{3}$$

And at $x = (3/2)L$

$$\frac{dv}{dx} = \frac{PL^2}{EI} \left[-\frac{17}{3} \left(\frac{3}{2}\right)^2 + \frac{7}{2} \left(\frac{3}{2} - 1\right)^2 + \frac{10}{6} \left(\frac{3}{2} - 1\right)^3 + \frac{35}{3} \right] = 0 \text{ (maximum)}$$

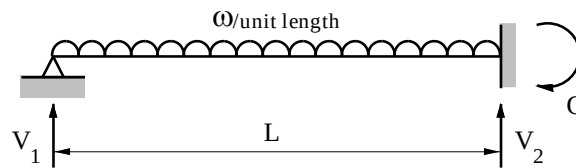
To find the actual deflection, use the equation of deflection

$$EIv = -\frac{17P}{9}x^3 + \frac{7P}{6}\langle x-L \rangle^3 + \frac{10P}{24L}\langle x-L \rangle^4 + \frac{35PL^2}{3}x$$

At this distance $3/2L$, the maximum deflection is:

$$v_{max} = \delta = \frac{PL^3}{EI} \left[\frac{-17}{9} \left(\frac{3}{2}\right)^3 + \frac{7}{6} \left(\frac{3}{2} - 1\right)^3 + \frac{10}{24} \left(\frac{3}{2} - 1\right)^4 + \frac{35}{3} \left(\frac{3}{2}\right) \right] = \frac{723 PL^3}{64 EI}$$

6



Equilibrium: $V_1 + V_2 = \omega L$
 $V_1 L + C = \omega L^2/2$

Elasticity: $\delta_1 = \frac{\omega L^4}{3EI} - \frac{V_1 L^3}{3EI}$

Compatibility: $\delta_1 = 0 \Rightarrow \frac{V_1 L^3}{3EI} = \frac{\omega L^4}{3EI} \therefore V_1 = \frac{3\omega L}{8}$

Hence $V_2 = \frac{5\omega L}{8}$ $C = \frac{\omega L^2}{8}$

